Multivariate Wind Speed Forecasting with Mutual Information Screening and Convex Ensemble Weighting

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Abstract

Wind speed forecasting is crucial for wind farm operations and power-system scheduling. However, forecasting pipelines can suffer from data leakage when feature selection or ensemble fitting uses evaluation data, and point forecasts alone provide limited information about uncertainty. To address these problems, we propose an auditable mutual-information (MI) screening and convex ensemble framework. We first screen features and fit scaling parameters on training-only data, then select base-model checkpoints on the outer validation block and calibrate nonnegative, unit-sum weights on the inner validation block using sequential least squares programming (SLSQP). We freeze the weights before evaluating the chronological test block. The framework separates the roles of the fitting, selection, calibration, and evaluation blocks and uses a moving-block bootstrap to produce intervals whose widths adapt to recent observed wind variability. Across the test block, the ensemble achieves a wind-speed mean absolute percentage error of 21.59% and an empirical prediction interval coverage probability of 95.70% at the nominal 95% level. These findings support further evaluation of the framework as an input to risk-aware wind farm scheduling.

Keywords: Wind speed forecasting, Mutual information, Convex ensembles, Sequential least squares programming, Chronological evaluation

1. Introduction

During the California heat wave of 15 August 2020, wind generation fell by about 1,200 MW in one hour. As reserves became insufficient, the system operator ordered approximately 500 MW of controlled load shedding for 20 minutes. The joint investigation identified several interacting causes, including extreme demand, inadequate evening resource availability, and market practices [1]. This event illustrates the operational stakes of anticipating sharp changes in available generation. In a separate electricity-market study, Weber and Woerman found that wind generation forecast errors imposed roughly twice the marginal operating cost impact of equivalent correctly forecast wind variation [2]. These observations motivate accurate short-horizon wind speed forecasts and uncertainty estimates for use alongside turbine power curves and operational constraints.

We study one-hour-ahead forecasting of wind speed jointly with five meteorological variables. The main design challenge is to combine predictors with different error patterns while assigning a clear role to each chronological data block. We use MI to screen candidate inputs, fit seven base predictors, and estimate a convex combination on an internal calibration

block. A moving-block bootstrap then converts calibration residuals into prediction intervals whose widths respond to recent observed wind variability.

This study makes three contributions.

- We specify an auditable chronological protocol that separates training-only MI screening and scaling, checkpoint selection, ensemble calibration, and final test evaluation.
- We calibrate a nonnegative, unit-sum ensemble over seven architectures and compare it with each base model and equal weighting using aggregate and wind-speed-specific metrics.
- We construct volatility-adaptive wind-speed prediction intervals using moving-block bootstrap calibration and evaluate empirical coverage and width alongside point forecast accuracy.

Section 2 reviews the relevant literature. The forecasting task and methods follow, then the experimental design and results. The discussion examines repeatability across training seeds and four contiguous test blocks. The conclusion summarizes the findings, and The appendices report the controlled feature and window experiments, paired error diagnostics, and reproduction details.

2. Related Work

2.1. Feature Selection

2.1.1. Physical screening and linear relevance

Physical knowledge can narrow the candidate set. Lagomarsino-Oneto et al. integrate physical guidance into machine-learning wind-speed prediction [3]. The physical structure helps delimit the functions that the model must learn from observations, but its usefulness depends on the representativeness of the measured quantities and the assumptions used to relate them. In particular, an explanatory physical variable is usable at deployment only if its value or an operational forecast of it is available at issuance. Linear association scores offer a complementary, inexpensive diagnostic. They can distinguish the sign of a linear association, but a large score can reflect a common diurnal cycle, and a small score can conceal a strong nonlinear dependence. Wind direction presents a further problem: 359° and 1° are close on a circle even though their scalar difference is large. A physically informed shortlist and a statistical ranking therefore address different parts of the input-selection problem.

2.1.2. Nonlinear and multivariate screening

Jiang et al. incorporate pairwise atmospheric dependencies into multistep wind-speed forecasting [4]. Such a representation can retain relationships that a separate score for each variable would omit, at the cost of estimating and maintaining a larger dependence structure. Wu et al. construct an expanded meteorological and statistical input set, apply two stages of selection, and train a temporal fusion Transformer with optimized hyperparameters [5]. Their use of TreeSHAP gives a way to inspect the selected inputs and fitted predictions. The seasonal evaluations at Williams Wind Farm also show that performance varies across periods under the same overall design. Because feature construction, selection, and predictor

optimization are combined, a matched ablation is required to determine how much of a gain can be attributed to screening alone. A more compact alternative is nearest-neighbour estimation of mutual information (MI) [6]. This detects nonlinear marginal dependence without committing to a linear response, but ranks variables independently. It can select redundant variables, omit interaction-only predictors, confuse contemporaneous relevance with forecast relevance, and yield estimates sensitive to the fitting period. These limitations motivate fitting MI on a dedicated chronological block and assessing the selected subset against a matched all-input arm.

2.2. Base Models

2.2.1. Statistical, recurrent, and decomposition-based predictors

Xu et al. combine seasonal-trend decomposition with statistical and attention-based models for offshore wind-speed assessment [7]. The comparison is informative because explicit seasonal structure can make simple extrapolation competitive, especially where recurrent daily patterns dominate. Its applicability to an hourly operational pipeline depends on whether the decomposition can be reproduced using only observations already available at each forecast origin. Lv and Wang couple hybrid decomposition with deep predictors and multi-objective tuning [8], while Liu and Wang combine training-set extension with a sequence integration architecture for very long forecast trajectories [9]. Both approaches broaden the model's access to temporal structure, but the effect of any one component is hard to isolate without matched variants. An NWP-assisted model can instead provide information about an approaching weather system before it appears in the local history [10]. That option depends on access to archived forecasts with documented issue times. Our comparison focuses on a setting where the local historical window is the common information source.

2.2.2. Attention and alternative temporal inductive biases

Wu et al. apply Transformer-based multistep forecasting directly to wind speed [11]. Jiang et al. examine Transformer applicability in a multivariate atmospheric framework [12]. Their design supports use of cross-variable information, although a model that combines learned temporal interactions with feature processing requires component-level comparisons before gains can be attributed to attention. Attention architectures have also been evaluated for power output [13], where turbine conversion and curtailment can make the target behave differently from speed. In general-purpose forecasting, iTransformer embeds each variate as a token, PatchTST groups contiguous observations into temporal patches, TimesNet represents intra- and interperiod variation, and Autoformer decomposes the series within the predictor [14, 15, 16, 17, 18]. Their different inductive biases make them informative candidates for combination. LSTM and DLinear provide recurrent and lightweight linear comparators [19, 20]. Model rankings may depend on forecast horizon, access to covariates, training budget, and temporal domain shift [21]. We compare these architectures on aligned timestamps, and interpret the ranking as conditional on their configurations and common forecast task.

2.3. Ensemble Methods

2.3.1. Combining subseries, predictors, and predictive distributions

Duan et al. combine convolutional and recurrent forecasting with regression-based correction that uses error information [22]. The correction stage can exploit systematic residual structure, but it introduces a separate fitting step whose inputs must be defined in chronological order. CEEMDAN–EWT hybrids apply different operations to decomposed signals [23]. Their forecasts combine representations of one signal, a different source of diversity from several architectures predicting the same target. Du et al. address interpretability explicitly within a wind-speed combined learning model [24]. The interpretability of a fitted combination is most useful when accompanied by information about the stability of its component weights. Takara et al. use stacking to combine advanced neural predictors with a probabilistic layer for multistep wind-power forecasting [25]. Stacking can capture interactions among base predictions but requires enough independent information to train the additional layer. The short calibration block in this study favors a lower-dimensional combination with coefficients shared across channels.

2.3.2. Chronological calibration and changing error dependence

Wang et al. explicitly address information leakage in a parallel differential learning ensemble for ultra-short-term wind-speed forecasting [26]. This work is relevant to feature transformations whose output at a given timestamp can depend on later input samples. A separate risk arises when an ensemble is estimated from predictions on the same targets used to score it. A well-defined evaluation therefore records the time support of each input transform, the labels used to select checkpoints, and the predictions and labels used to calibrate the combiner. Sun et al. report that the preferred decomposition-based forecasting pipeline varies with site and lead time [27]. By analogy, a fixed set of combination weights can be effective when residual relationships persist and less effective when those relationships change. Equal weighting avoids estimation variance but gives the same influence to predictors with different bias or error dependence. Flexible stacking can reduce approximation error but requires a larger calibration set. Fixed convex weights keep the calibration auditable, and can still be sensitive to correlated errors, calibration size, model availability, and temporal shifts. Our analysis tests these trade-offs through comparisons with equal weighting and the selected single model.

2.4. Interval Prediction

2.4.1. Distributional and jointly trained intervals

Baggio and Muzy use an M-Rice predictive distribution and spatial observations to represent wind-speed uncertainty [28]. A parametric distribution yields coherent quantiles and a natural treatment of wind-speed support, while inference depends on the adequacy of the chosen family in the tails and the availability of spatial measurements. Chen et al. use dynamic spatio-temporal graph convolution to forecast wind-speed intervals [29]. An evolving graph can reflect shifting spatial relationships, but requires enough synchronized stations to estimate those relationships. Chen et al. subsequently combine adaptive Bayesian wavelet denoising with deep Gaussian process regression and compare results across four wind farms [30]. Denoising and a probabilistic predictor offer a way to separate input variability from

predictive uncertainty. For a sharp ramp, however, the operational consequence of denoising depends on whether the removed component was noise or a genuine change. Saeed et al. train a branched LSTM ensemble to estimate wind-speed point forecasts and intervals jointly [31]. A joint objective can adapt to the conditional distribution, but the balance between sharpness and empirical coverage can depend on the training loss and sampling of rare conditions.

2.4.2. Residual calibration and temporal dependence

Quantile forecasting, conformal calibration, kernel density estimation, and distributional modelling for extreme events provide alternative paths to predictive bounds [32, 33, 34, 35, 36, 37]. A residual approach permits recalibration around a frozen point model and preserves the separation between point estimation and the construction of uncertainty bounds. Resampling individual residuals as if independent disregards the fact that errors can cluster across consecutive forecast origins. Moving-block resampling preserves within-block sequences, while a scale estimated from trailing observations allows the bounds to widen when recent conditions become more variable. The calibration still depends on the chosen block length, the adequacy of the local scale, the representativeness of the residual period, and the rarity of abrupt changes. Pointwise coverage averaged over time can also conceal concentrations of misses. We report both coverage and width and specify the calibration information available to the operational forecaster. Table 3 summarizes the methodological scope and trade-offs of 15 recent wind-speed studies. Differences in site, horizon, and outcome preclude direct comparison of their numerical scores.

3. Forecasting task

Let $\boldsymbol{x}_t \in \mathbb{R}^C$ denote the observations available at time t. Model m maps the window $\boldsymbol{X}_t = (\boldsymbol{x}_{t-L+1}, \ldots, \boldsymbol{x}_t)$ to $\widehat{\boldsymbol{Y}}_t^{(m)} \in \mathbb{R}^{H\times C}$. We set $L = 24$, $H = 1$, and $C = 6$. The outputs comprise five meteorological variables and wind speed. Each forecast uses observations through its origin and the known target timestamp. Forecast windows may cross a split boundary to use earlier observations, while target timestamps belong to disjoint blocks. During the test period, earlier observations may be used as they become available for the next one-hour-ahead forecast.

4. Methods

4.1. Mutual information screening

For a candidate variable X_j and contemporaneous wind speed Y, MI is

$$I(X_j;Y) = \iint p_{X_j,Y}(x,y)\log\frac{p_{X_j,Y}(x,y)}{p_{X_j}(x)p_Y(y)}\,\mathrm{d}x\,\mathrm{d}y. \tag{1}$$

We estimate MI on the first 1,060 records using the nearest-neighbour estimator implemented in `mutual_info_regression` [6], with three neighbours and random seed 42. We prespecify five covariates and retain wind speed as an additional input. The selected variables, in descending MI order, are wind direction, solar radiation, air temperature, relative humidity,

and reference evapotranspiration. The contemporaneous score measures association within the fitting block, whereas prediction uses historical observations. Marginal MI may retain redundant variables or miss variables informative only through interactions. Wind direction is represented numerically in this experiment. Circular encoding and conditional screening remain relevant controls.

4.2. Base predictors

LSTM uses gated recurrence followed by a direct output projection [19]. Transformer uses an attention-based encoder and decoder [14]. iTransformer treats variates as tokens to model cross-variable dependencies [15]. PatchTST constructs representations from temporal patches using a channel-independent design [16]. TimesNet organizes one-dimensional signals into two-dimensional representations of intra- and interperiod variation [17]. DLinear applies linear forecasting heads to trend and seasonal components [20]. Autoformer incorporates progressive decomposition and an autocorrelation mechanism [18]. All predictors use the same observation windows, target order, and chronological blocks.

4.3. Convex ensemble calibration

Let $P \in \mathbb{R}^{n\times M}$ contain aligned predictions on Inner-Val, where $M = 7$, $n = N_{\text{in}}HC$, and each column stacks the predictions of one model. Let $\boldsymbol{y}$ stack the corresponding targets. The admissible weights lie in the simplex

$$\Delta_M = \{\boldsymbol{w} \in \mathbb{R}^M : w_m \geq 0,\ \mathbf{1}^\mathsf{T}\boldsymbol{w} = 1\}. \tag{2}$$

The calibrated weights are

$$\boldsymbol{w}^* = \arg\min_{\boldsymbol{w}\in\Delta_M} \frac{1}{n}\|P\boldsymbol{w} - \boldsymbol{y}\|_2^2. \tag{3}$$

Let row m of E contain the residuals of model m, and define $G = EE^\mathsf{T}/n$. Since the weights sum to one, the objective equals $J(\boldsymbol{w}) = \boldsymbol{w}^\mathsf{T}G\boldsymbol{w}$ on the feasible set. Its gradient for this quadratic representation is $2G\boldsymbol{w}$ and its Hessian is $2G \succeq 0$. The cross terms in G capture the extent to which the model errors reinforce or offset one another. The simplex permits zero weights and confines each ensemble forecast to the range spanned by the base forecasts. An optimum need not be unique when residual directions are linearly dependent.

We use SciPy's sequential least squares programming (SLSQP) implementation [38], initialize at $w_m = 1/M$, supply the analytic gradient, and set the stopping tolerance to 10^{-12} and the iteration limit to 2,000. We check convergence and the simplex constraints. The equal-weight ensemble uses $w_m = 1/M$ throughout.

Solver choice and numerical verification.. Equation (3) is a seven-variable convex quadratic program with an analytic gradient and linear constraints. SLSQP provides direct access to this gradient and the simplex constraints through SciPy's general constrained-optimization interface. We cross-checked the solution with OSQP [39], using the same Inner-Val Gram matrix and comparing the two feasible solutions after correcting roundoff at the simplex boundary. Over 100 fresh solves of the fixed problem, with solver setup included and shared Gram-matrix construction excluded, the CPU median (99th-percentile) time was

0.726 (1.138) ms for SLSQP (SciPy 1.16.3) and 1.074 (1.505) ms for OSQP 1.0.4. These timings describe this small instance and the stated Python implementations. The maximum absolute difference in weights was 1.77×10^{-7}, the absolute difference in Inner-Val objective values was 1.61×10^{-15}, and the largest absolute difference between their frozen standardized test predictions was 2.52×10^{-7}. The agreement supports the numerical adequacy of SLSQP for this instance. The contribution lies in the chronological calibration protocol and the constrained combination, with SLSQP serving as the implementation used throughout.

5. Experimental design

5.1. Data and chronological separation

The supplied cleaned series contains 2,208 consecutive hourly records from 1 November 2023 at 01:00 to 1 February 2024 at 00:00. We use the recorded timestamps. The outer partition allocates 1,324 records to the development block and 442 each to Outer-Val and the test block. Within the development block, the first 1,060 records form the fitting block and the final 264 form Inner-Val. The resulting half-open, zero-based ranges are $[0, 1060)$, $[1060, 1324)$, $[1324, 1766)$, and $[1766, 2208)$. The first 24 fitting records provide context, leaving 1,036 fitting windows.

The feature ranking and standardization statistics use the fitting block alone. Outer-Val determines early stopping and the minimum-MSE checkpoint for each model. Predictions from those checkpoints on Inner-Val calibrate the ensemble weights and identify the single-model comparator by Inner-Val MSE. The weights are written to disk before constructing test inputs and generating test predictions. Outer-Val occurs later than Inner-Val in calendar time, so the Inner-Val errors are calibration errors conditional on checkpoint selection. The final test block follows both validation blocks.

5.2. Training configuration

Each model is initialized with seed 2021 and trained with Adam at a fixed learning rate of 10^{-4}, a batch size of 16, and a maximum of 30 epochs. Training stops after five consecutive epochs without an improvement in Outer-Val MSE, and the best checkpoint is restored. The decoder label length is 12. Forecast inputs at unknown future positions are zero-padded. Evaluation uses float32 and a batch size of one to prevent batch-dependent operations, such as TimesNet's frequency selection, from coupling forecasts at different origins.

The hidden size is 512 for Transformer, iTransformer, and LSTM, and 64 for PatchTST, TimesNet, and Autoformer. The corresponding feed-forward widths where applicable are 2,048 and 128. The configuration specifies two encoder or recurrent layers, one decoder layer where applicable, eight attention heads, and dropout of 0.1 where supported. DLinear and Autoformer use a moving-average kernel of 25. Model sizes differ across architectures.

5.3. Evaluation metrics

For N test samples and C standardized outputs, we report

$$\mathrm{MSE} = \frac{1}{NC}\sum_{i,c}(y_{ic} - \hat{y}_{ic})^2, \tag{4}$$

$$\mathrm{MAE} = \frac{1}{NC}\sum_{i,c}|y_{ic} - \hat{y}_{ic}|, \tag{5}$$

$$R^2_{\mathrm{avg}} = \frac{1}{C}\sum_{c=1}^{C}\left(1 - \frac{\sum_i (y_{ic} - \hat{y}_{ic})^2}{\sum_i (y_{ic} - \bar{y}_c)^2}\right). \tag{6}$$

For wind speed alone, we also report errors after inverse standardization using the fitting-block mean and scale. Wind-speed MSE and MAE are expressed in mph^2 and mph, respectively. Applying the same affine transformation to targets and forecasts leaves R^2 invariant.

6. Results

6.1. Point Forecasting Results

Table 1 compares seven single models and two ensembles. Transformer has the lowest aggregate MSE among individual models and is also the comparator selected on Inner-Val. The optimized ensemble reduces aggregate MSE by 5.49% relative to Transformer and by 3.33% relative to equal weighting. For wind speed, the optimized ensemble attains an R^2 of 0.681449, compared with 0.670007 for Transformer and 0.653769 for equal weighting. In standardized units, its wind-speed MSE and MAE are 0.249186 and 0.350875, respectively. These channel-specific outcomes are reported separately from the six-output averages.

Table 1: Performance on the final 442 test observations. Aggregate errors are computed on standardized outputs and aggregate R^2 is the unweighted mean of the six channel scores. Wind-speed MSE and MAE use mph^2 and mph, respectively.

	Six standardized outputs			Wind speed		
Method	MSE	MAE	Mean R^2	MSE	MAE	R^2
iTransformer	0.240319	0.277415	0.699330	0.650078	0.562275	0.624377
PatchTST	0.240720	0.268708	0.700059	0.663020	0.566494	0.616898
TimesNet	0.235966	0.281516	0.701629	0.663663	0.582011	0.616527
LSTM	0.248825	0.308422	0.658681	0.765975	0.628452	0.557409
Transformer	0.215200	0.287707	0.718794	0.571107	0.534422	0.670007
DLinear	0.251103	0.304394	0.669874	0.746943	0.610856	0.568406
Autoformer	0.333511	0.393732	0.540085	0.998197	0.765554	0.423229
Equal weight	0.210406	0.262023	0.734235	0.599209	0.552030	0.653769
SLSQP	**0.203396**	**0.259194**	**0.744992**	**0.551304**	**0.521900**	**0.681449**

6.2. Ensemble Weights and Residual Diagnostics

Table 2 reports the frozen weights. Transformer, iTransformer, and TimesNet form the active set. PatchTST, DLinear, Autoformer, and LSTM receive zero weight to the precision shown. In the simplex-constrained problem, a boundary solution can arise when reallocating mass from the active mixture to an inactive predictor cannot reduce the calibration objective locally. The objective depends on joint residuals, so a competitive standalone model can receive zero weight if its errors add little complementarity. Sparsity prevents such models from diluting the fitted combination on Inner-Val. Its effect on future periods depends on the stability of those residual relationships.

Table 2: Frozen ensemble weights estimated on Inner-Val.

Model	Weight (%)
Transformer	51.83
iTransformer	24.83
TimesNet	23.34
PatchTST	0.00
DLinear	0.00
Autoformer	0.00
LSTM	0.00

The comparison with equal weighting over all seven candidates varies both the set of models contributing to the forecast and the relative weights of those models. A direct control in future work will assign weights of 1/3 to Transformer, iTransformer, and TimesNet, the three materially weighted models in Table 2, and zero to the other candidates. Using identical base forecasts, forecast origins, and evaluation metrics for this control and SLSQP will hold ensemble membership and size fixed. The resulting contrast would estimate the contribution of unequal weighting conditional on this active set.

The SLSQP ensemble attains the lowest full-test wind-speed MAPE among the nine methods at 21.59%, compared with 21.78% for Transformer and 23.72% for equal weighting. The five equally spaced axes in Fig. 1(d) share a common radial scale. The SLSQP polygon has the smallest area under that construction. The ranking varies by block: SLSQP has lower MAPE than Transformer in blocks 2, 4, and 5, while Transformer performs better in blocks 1 and 3. Polygon area summarizes this particular set of axes and is sensitive to axis ordering. The full-test MAPE provides the primary numerical comparison.

The shaded interval in Fig. 1(a) identifies observations 68–80. We selected it using observed wind-speed changes alone, independently of the models' errors. In this window, Transformer has an MAE of 0.801 mph and SLSQP has an MAE of 0.815 mph. Both forecasts lag the sharp rise, which also produces the largest SLSQP error in the displayed sequence. The figure therefore documents a local exception to the aggregate advantage.

6.2.1. Wind-speed error by observed regime and member removal

We stratified the saved test forecasts by observed wind speed and absolute one-hour wind-speed change. Thresholds were obtained from the fitting block: the median speed was 2.4 mph and the 75th percentile of absolute hourly change was 0.7 mph. We first assigned

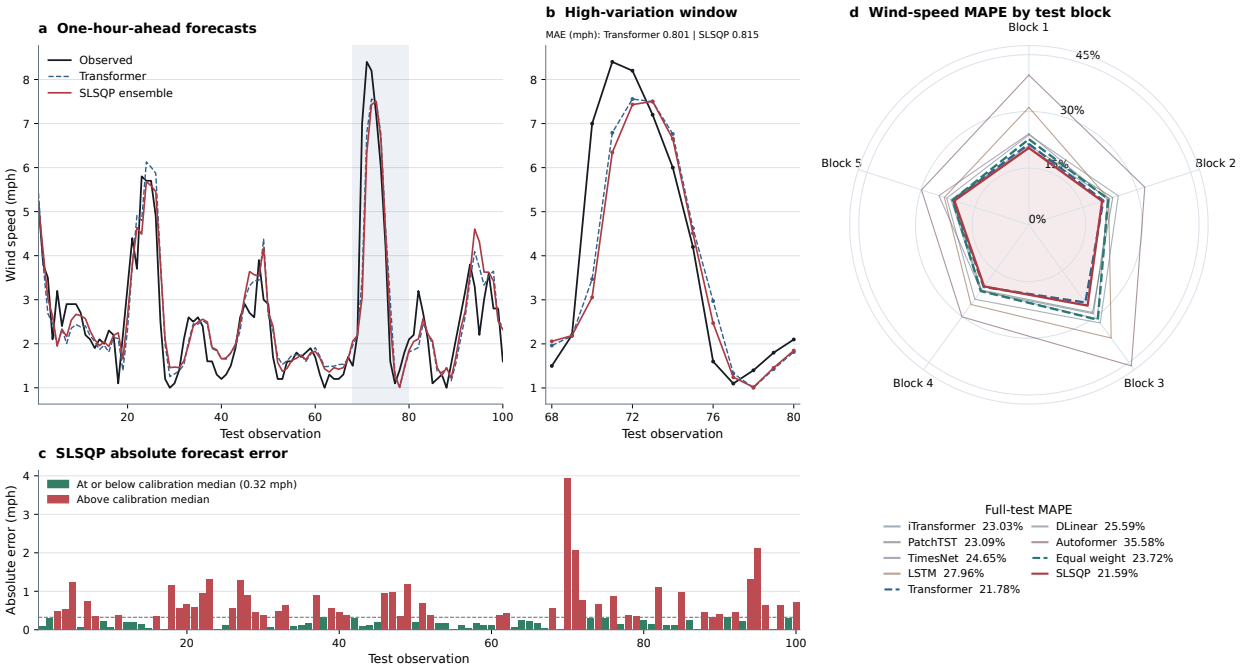


Figure 1: Point forecasts for wind speed. (a) Observed values and one-hour-ahead predictions for the first 100 test observations. (b) The 13-observation window with the largest mean absolute change in observed wind speed. (c) Absolute SLSQP errors, coloured by whether they exceed the median absolute error on Inner-Val. (d) MAPE across five contiguous test blocks, with full-test MAPE in the legend.

changes above 0.7 mph to the high-ramp group. Of the remaining targets, speeds at or below 2.4 mph were assigned to the low-wind group and all others to the moderate-change group. We use the last Outer-Val observation to compute the first test change. The grouping is a retrospective diagnostic based on realized conditions; it is not an input to the forecasts. An hourly speed change measures a ramp at the sampling resolution and cannot establish the presence of a sub-hourly gust.

In Table 4, Transformer has the lowest standalone RMSE in the high-ramp group (1.2235 mph), and the frozen ensemble improves on it (1.2076 mph). To assess contributions conditional on the remaining candidates, we removed each materially weighted model in turn, refitted the simplex weights over the other six saved Inner-Val forecasts, and applied those weights to the unchanged test forecasts. No base model was retrained. Removing Transformer yields the largest increase in both aggregate MSE (0.011524) and high-ramp RMSE (0.02635 mph) among the three omissions. Its omission also produces the largest deterioration in low wind, so its contribution is not confined to rapid changes.

In the moderate-change group, the full ensemble improves on all three standalone predictors, but removing iTransformer and reallocating its weight reduces RMSE by 0.00774 mph. Its omission increases aggregate MSE and high-ramp RMSE, so the evidence for its marginal contribution depends on the channel and regime. Removing TimesNet raises moderate-change RMSE by 0.02370 mph while slightly reducing high-ramp RMSE and pooled wind MSE. These patterns are consistent with different error responses to observed changes in the wind series. They do not identify which atmospheric processes individual models represent.

The weights optimize six standardized outputs jointly, and the omission effects also reflect the availability of replacement models in the candidate pool.

6.3. Interval Forecasting Results

We construct nominal 95% wind-speed prediction intervals on the same 442-observation test block as the point forecast evaluation. For target index t, let v_t be the root mean square of the hourly changes over the previous 12 observed wind speeds, from $t-12$ through $t-1$. We use the local scale $s_t = \max(v_t, s_{\min})$, where $s_{\min}$ is the 25th percentile of historical volatility computed in the fitting block. In this experiment the floor is 0.4906 mph.

We divide the 264 signed SLSQP residuals on Inner-Val by their corresponding s_t values. Each of 2,000 moving-block bootstrap replicates samples contiguous 12-observation blocks with replacement and truncates the concatenation to the calibration length. We use random seed 42. We average the replicate-specific 2.5th and 97.5th percentiles to obtain lower and upper residual offsets q_L and q_U. The bounds for a test forecast $\hat{y}_t$ are

$$L_t = \max(0, \hat{y}_t + s_t q_L), \qquad U_t = \max(0, \hat{y}_t + s_t q_U). \tag{7}$$

Interval construction uses the frozen offsets and observations available before the target. Prior to clipping at zero, width increases linearly with the local scale.

The mean prediction interval width (MPIW), prediction interval coverage probability (PICP), and normalized average width (PINAW) are

$$\text{MPIW} = N^{-1} \sum_t (U_t - L_t), \tag{8}$$

$$\text{PICP} = N^{-1} \sum_t \mathbf{1}\{L_t \le y_t \le U_t\}, \tag{9}$$

$$\text{PINAW} = \frac{\text{MPIW}}{\max_t y_t - \min_t y_t}. \tag{10}$$

The observed test range is used only for width normalization at evaluation. Across all 442 test targets, the adaptive interval covers 423 observations (PICP 95.70%), with PINAW 0.4990 and MPIW 3.6926 mph. Widths range from 2.2805 to 7.6445 mph. In Fig. 2, black points denote actual observations and the translucent red band denotes the nominal 95% prediction interval. The lower panel shows how the band widens after a sharp change enters the trailing window and narrows as the observations settle.

Inner-Val also supplies the ensemble weights, so the bootstrap calibrates residuals from a combination fitted on that block. Moving blocks preserve local serial dependence in the resampled residuals. The reported PICP describes this test period, and calibration stability across regimes remains to be assessed.

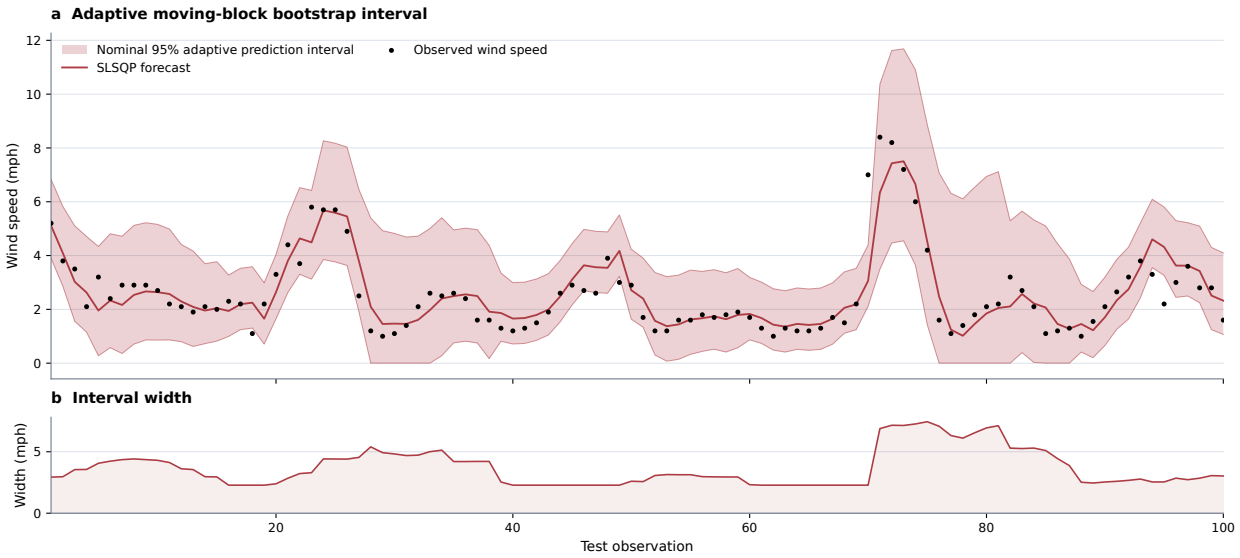


Figure 2: Adaptive wind-speed prediction intervals for the first 100 test observations. (a) Observations (black points), SLSQP point forecasts (red line), and nominal 95% moving-block bootstrap intervals (red shading). (b) Interval width in mph. Coverage and width statistics in the text use the complete test block.

7. Discussion

The supplementary controls in Appendix A examine the two main input-design decisions. In iTransformer, the MI subset reduces test aggregate MSE from 0.243347 to 0.240319 under a matched output definition. The history-length sweep selects 24 hours by Outer-Val MSE, while 96 hours yields the lowest aggregate test MSE. We retain the validation-based choice in the main pipeline and use the test sweep to characterize sensitivity.

7.1. Accuracy across outputs, model classes, and time blocks

Table 1 distinguishes the six-output optimization target from wind-speed performance in physical units. Under the same chronological split, the linear DLinear baseline attains an aggregate MSE of 0.251103, compared with 0.215200 for Transformer and 0.203396 for the SLSQP ensemble. These results describe the specified configurations and training budget. They do not establish a general ranking of linear and nonlinear methods. The ensemble leads both aggregate and wind-specific metrics in this experiment, yet the block and zoom comparisons in Fig. 1 reveal variation in relative performance. The observed wind-speed MAPE across the five blocks ranges from 20.14% to 26.42% for SLSQP and from 20.34% to 25.40% for Transformer. Thus the smaller aggregate error coexists with harder blocks and a local reversal during a sharp ramp.

7.2. Protocol and evidence limitations

The experiment starts from the supplied cleaned series. The original station, licensing, and preprocessing history require documentation, particularly for any smoothing or interpolation across time boundaries. Earlier exploratory runs inspected timestamps that substantially overlap the present test block. Given the chronology of the study, the current

test constitutes a retrospective evaluation. A genuinely blinded assessment requires independent observations that have not been inspected during development and are evaluated only after the protocol has been frozen. Checkpoint selection uses Outer-Val, which is later than Inner-Val, and the latter also supplies interval calibration residuals. The seed and time-block analyses in Sections 7.3 and 7.4 examine variation within the available record. An independent later period or external station would support stronger conclusions on generalization and interval stability.

The experiment also changes several design elements relative to earlier runs, including split boundaries and candidate models. Appendix A.1 compares all inputs with training-only MI screening. A matched-count Pearson comparator would further isolate the choice of screening criterion. Appendix D provides a persistence reference for wind speed. Matched model budgets would sharpen interpretation of the architecture comparison. Dispatch-level evidence would require mapping speed forecasts to turbine power and evaluating an explicit scheduling policy.

7.3. Multi-Seed Stability

We reran the seven base models using seeds 2021, 42, and 2023. Each run used the same chronological split, feature-screening rule, training budget, checkpoint criterion, and test timestamps. Base forecasts and SLSQP weights were regenerated for each seed, with weight calibration confined to Inner-Val. This setup examines the repeatability of the complete point-forecast pipeline under changes in training randomness, including model initialization and the order of training batches. The MI estimator used its fixed seed of 42 in all three runs. Keeping that seed fixed holds the definition of the six-output task constant.

The standardized test MSEs were 0.203396, 0.204907, and 0.202490 for seeds 2021, 42, and 2023, respectively. The corresponding six-output mean R^2 values were 0.744992, 0.740994, and 0.746906. The sample mean and standard deviation of MSE were 0.203598 ± 0.001221 ($n = 3$), as shown in Fig. 3. The standard deviation is about 0.60% of the mean and the full observed range is 0.002417. The principal run (seed 2021) lies within that range and close to its centre. All three ensemble MSEs are below the 0.215200 of the seed-2021 Transformer reference. This cross-run comparison is descriptive because the reference is a single realization. A paired comparison across a larger set of seeds would more directly quantify the seed dependence of the ensemble's gain.

The wind-speed channel exhibits more variation than the aggregate score. Its mean R^2 across seeds is 0.666098 ± 0.013987, and its mean physical-unit MSE is 0.577873 ± 0.024207 mph^2. Seed 2023 gives the lowest aggregate MSE but the highest wind-speed MSE, 0.598679 mph^2. Conversely, seed 2021 has the lowest wind-speed MSE, 0.551304 mph^2. The two rankings arise from different loss functions: aggregate squared error pools the six standardized channels, whereas the wind-speed loss singles out one channel. When the forecast feeds a wind-specific operational decision, aggregate repeatability cannot substitute for direct assessment of that channel.

Transformer receives the largest weight under all three seeds, while iTransformer and TimesNet each remain substantive contributors. Their relative shares change across runs, and Autoformer receives a small positive weight under two seeds (Table B.7). The combination is thus similar in composition across the tested seeds even though individual coefficients are sensitive to model training. If two fitted predictors make nearly identical forecasts, weight

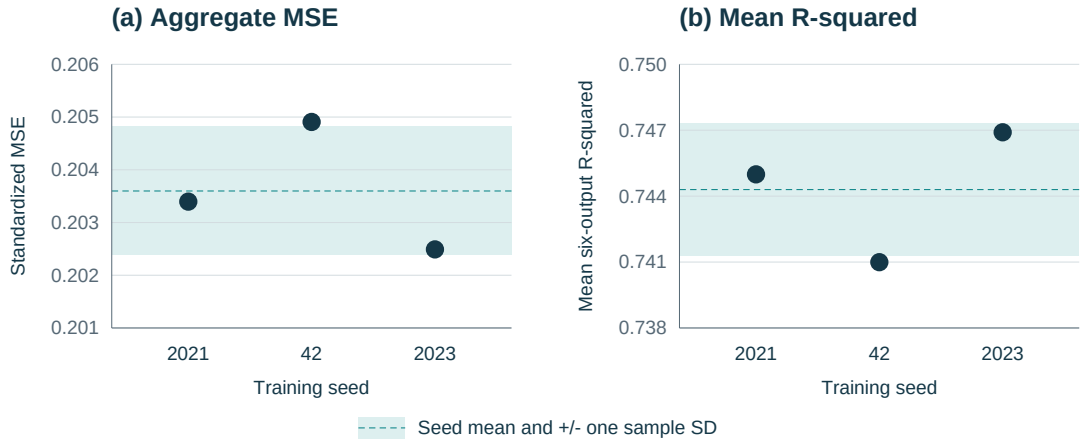


Figure 3: Standardized six-output test MSE and mean R^2 across three independent training runs on the fixed chronological split. The horizontal line is the seed mean and the shaded band spans one sample standard deviation on each side. Each point represents a full retraining and recalibration.

redistribution between them can change coefficients appreciably while barely affecting the weighted prediction. Conversely, a fixed weight vector can produce different wind errors if a base model changes the timing or amplitude of its forecast. It is therefore useful to report both the point-metric dispersion and the active composition. The three-run dispersion is a sample statistic for training variation on a fixed dataset. It cannot resolve rare bad initializations, uncertainty due to time sampling, or sensitivity to the choice of station.

7.4. Temporal Robustness

We divided the 442 hourly test targets into four contiguous periods containing 111, 111, 110, and 110 observations. Each method issued forecasts under the same frozen configuration in all four periods. We computed wind-speed MAPE from the inverse-transformed forecasts and observations. The diagnostic excludes observations with absolute speed at or below 10^{-6} mph, and none of the test observations were excluded. The four segments span about 18 days in January and early February; they represent adjacent blocks of one test record. The five segments used for the radar plot in Fig. 1 use a different partition.

Figure 4 and Table B.8 show that SLSQP improves on Transformer by 0.74 and 0.29 percentage points in Periods 1 and 2. Transformer is ahead by 0.20 and 0.08 percentage points in Periods 3 and 4. When the 442 targets are pooled, SLSQP has 21.59% MAPE versus 21.78% for Transformer. Its two earlier gains outweigh its two later losses. Equal weighting has the largest MAPE in each block, exceeding SLSQP by 0.63–3.86 percentage points; the largest gap is in Period 3. Equal weighting assigns four of seven shares to models that receive effectively zero weight in the principal run. This illustrates the cost of allocating a large fraction of the forecast to predictors that contribute little to the calibrated quadratic objective. The observation applies to the available model pool and split. Equal weighting remains a relevant reference because it uses no calibration targets.

We also examined the observed winds and the unrounded model errors (Table B.8). The mean wind speed is lowest in Period 3, at 2.25 mph, and its within-period root-mean-square

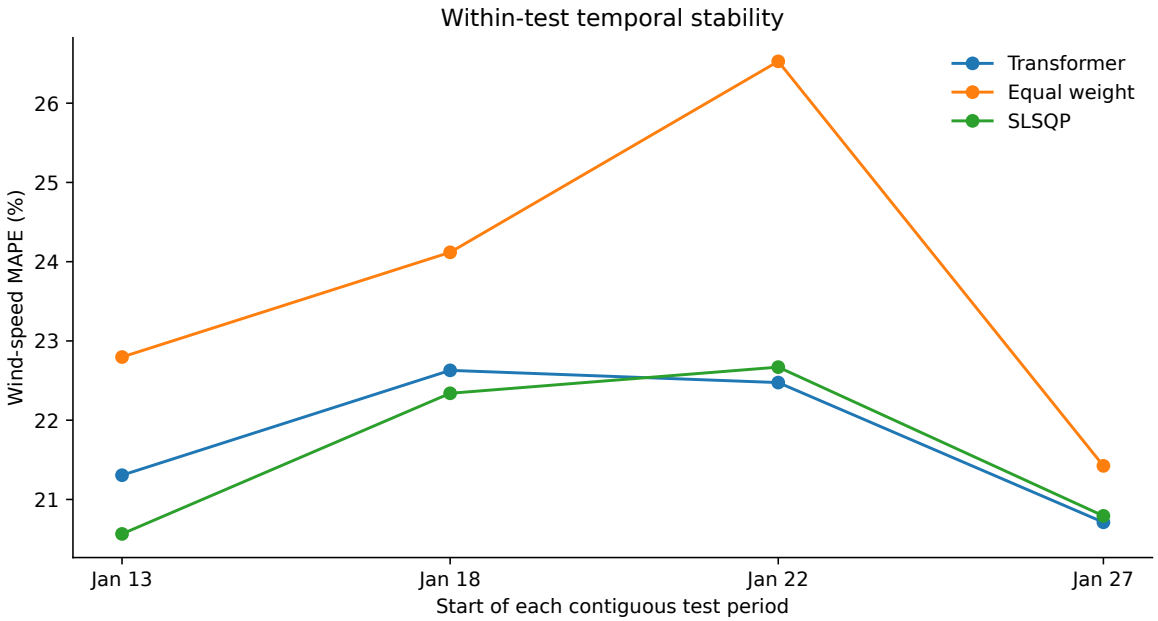


Figure 4: Wind-speed MAPE for SLSQP, Transformer, and equal weighting in four contiguous test periods. The labels identify the starting dates in 2024. Lines connect separate block summaries to show their ordering over time.

hourly change is 0.80 mph, the lowest of the four periods. For a fixed absolute speed error, MAPE assigns a greater penalty at low observed speeds. In Period 3, SLSQP has lower wind-speed MSE than Transformer (0.467 versus 0.492 mph^2) even though Transformer has the lower MAPE. Period 4 exhibits the largest hourly changes (1.02 mph RMS); here too, SLSQP has lower wind-speed MSE (0.682 versus 0.745 mph^2) but slightly higher MAPE. In Period 1, the ranking goes the other way: SLSQP reduces MAPE while raising wind-speed MSE. These contrasts show that the two methods distribute errors differently over the wind-speed range. They also caution against interpreting a reversal in MAPE as a reversal under an operational loss that penalizes large deviations more heavily.

Across the four periods, SLSQP has the lower absolute error at 58.6%, 53.2%, 49.1%, and 43.6% of forecast origins, respectively. The relative frequency of improvements declines even though the squared-error benefit grows from Period 3 to Period 4. The forecast differences among the active models can reduce some large errors while making more of the smaller errors slightly worse. This is consistent with complementary predictions under squared loss and a shift in the relative benefit across time. A physical interpretation is that local persistence, directional shifts, and changing boundary-layer conditions can alter the relative predictability of slow variations and short ramps. A convex average can damp overshoot when models disagree in amplitude; it can also lag a transition if the strongest single model responds earlier than the other active components. The measured speeds and forecast errors support the statistical account, but identifying a physical cause would require independent evidence such as upstream weather fields, boundary-layer observations, or event labels.

The temporal analysis uses one short winter record and three forecast methods trained on common data. Neighbouring periods can share weather systems, and individual errors are

serially dependent. The four blocks therefore describe within-test variation. For validation over seasons or sites, a prospective study would repeat the chronological protocol on independent records and report paired loss differences by pre-defined wind regime. The present comparison supports maintaining a Transformer reference alongside the fixed combination when monitoring deployment performance, particularly where the relevant objective differs from standardized aggregate MSE.

7.5. Implications for evaluation and operational use

An operator deciding how much flexibility to schedule would use the forecast in conjunction with the turbine power curve, current turbine availability, and operating constraints. The same wind-speed error can have markedly different energy consequences depending on the part of the power curve, and a point prediction can be economically satisfactory even when it misses an extreme. MAPE additionally places more emphasis on low measured speed, where absolute power production may be modest. In contrast, sustained errors near transitions in power output can matter even when their relative speed error is small. A direct economic assessment would fix a scheduling policy, propagate speed uncertainty through a power conversion model, and evaluate energy imbalance or reserve cost with archived operational data. The speed results in this paper provide a basis for that assessment.

The interval results add information about which point forecasts carry greater uncertainty. The trailing-volatility scale reacts after wind changes enter the observed history, so interval width can lag the first hour of a sudden event. This is an important deployment property for operators who care about the onset of ramps. A causal NWP input or upwind observation could supply earlier evidence, subject to its issue time and availability. In addition, marginal coverage can remain near 95% even if several consecutive misses occur during a difficult episode. Reporting the directions of interval misses, run lengths of misses, and coverage by an independently defined volatility stratum would test whether the intervals remain useful at the times of highest operational risk.

The experiments address four separate questions. The feature ablation tests whether reducing the input set helps one architecture on the chosen split. The window analysis tests the sensitivity of a model to history length under matched fitting timestamps. The seed analysis tests repeatability of the trained pipeline, and the four-block analysis tests stability of the relative forecast errors within the held-out window. Each answer has a different unit of analysis: an input configuration, a history length, a retraining run, or a sequence of test targets. Keeping these distinctions is important when deciding which parts of the pipeline to carry over to a new site. For example, small cross-seed dispersion is compatible with an unstable feature ranking under a new climate record. Likewise, a stable feature set is compatible with shifts in the cross-model residual covariance used for weighting.

8. Conclusion

The chronological experiment combines training-only MI screening with SLSQP weighting and volatility-adaptive prediction intervals. On the final test block, the ensemble attains a standardized aggregate MSE of 0.203396, a mean six-output R^2 of 0.744992, a wind-speed R^2 of 0.681449, and a wind-speed MAPE of 21.59%. The nominal 95% intervals achieve

95.70% empirical coverage with a PINAW of 0.4990. Across three training seeds the aggregate MSE is 0.203598 ± 0.001221. The four-period comparison shows smaller pooled wind MAPE for SLSQP and modest local advantages for Transformer in two periods. The matched iTransformer ablation favors the MI-screened inputs; the history-length sweep favors 24 hours by the validation criterion. The study therefore supports the use of separately calibrated convex weights as a repeatable point-forecast design on this record. The point forecasts and uncertainty estimates provide inputs for evaluating risk-aware wind farm scheduling under operational constraints.

Table 3: Comparison of recent wind speed forecasting studies

Category	Methods	Advantages	Disadvantages	Representative Literature
Physics-informed	Physical guidance with machine learning	Brings domain structure into the predictor	Depends on the relevance of physical assumptions and inputs	Lagomarsino-Oneto et al., *Energy* (2023) [3]
Statistical hybrid	STL with statistical and attention models	Separates seasonality and enables model comparison	Requires decomposition choices and site-level assessment	Xu et al., *Renewable Energy* (2023) [7]
Multivariate	VMD, graph modelling, and temporal forecasters	Represents cross-variable and multiscale structure	Adds preprocessing and graph configuration	Jiang et al., *Applied Energy* (2024) [12]
Combination	Decomposition, CNN–RNN, and regression	Combines subseries forecasts and error information	Requires fitting several components and a combiner	Duan et al., *Renewable Energy* (2022) [22]
Leakage control	Parallel differential ensemble with leakage mitigation	Addresses decomposition-related information leakage	Multi-stage design adds implementation complexity	Wang et al., *Applied Energy* (2024) [26]
Distribution	Spatial data and M-Rice predictive distribution	Produces a distribution with spatial context	Relies on a parametric family and spatial inputs	Baggio and Muzy, *Applied Energy* (2024) [28]
Interval	Dynamic spatio-temporal graph convolution	Incorporates spatial and temporal dependencies into bounds	Requires spatial observations and graph estimation	Chen et al., *Energy* (2024) [29]
Pairwise dependence	Multistep modelling of atmospheric dependencies	Retains cross-variable structure	Requires estimating dependencies and matching the forecast horizon	Jiang et al., *Energy* (2023) [4]
Nonlinear screening	Two-stage selection, TFT, and TreeSHAP	Combines input screening and attribution	Joint tuning complicates attribution of gains	Wu et al., *Renewable Energy* (2026) [5]
Deep hybrid	Multiscale decomposition and multi-objective tuning	Represents different temporal scales	Adds transformations and tuning stages	Lv and Wang, *Applied Energy* (2022) [8]

Table 3 continued

Category	Methods	Advantages	Disadvantages	Representative Literature
Attention	Transformer for multistep prediction	Learns temporal relationships directly	Skill can depend on lead and available context	Wu et al., *Energy* (2022) [11]
Interpretable ensemble	Combined learning with interpretability	Makes forecast behaviour easier to inspect	Explanations depend on fitted components and evaluation regime	Du et al., *Applied Energy* (2024) [24]
Signal splitting	Decomposition-based deterministic and probabilistic forecasts	Exposes differences across sites and horizons	A preferred pipeline may vary by location and lead	Sun et al., *Renewable Energy* (2025) [27]
Gaussian process	Bayesian wavelet denoising and deep Gaussian process	Couples denoising and predictive uncertainty	Denoising may attenuate genuine ramps	Chen et al., *Renewable Energy* (2026) [30]
Joint intervals	Branched LSTM with joint mean and variance loss	Produces point and interval estimates together	Sharpness and coverage depend on the fitted loss	Saeed et al., *Energy* (2025) [31]

Reproducibility

Training is implemented in `nested_retrain.py`. Point evaluation is implemented in `nested_evaluation.py`. The run directory `nested_60_20_20_run` records the split boundaries, scaling statistics, configurations, checkpoints, weights, and predictions. The post-hoc regime, member-removal, and solver diagnostics use `reviewer2_diagnostics.py`. The checkpoint benchmark uses `benchmark_frozen_inference.py`. The main point and interval figures and the adaptive interval metrics can be reproduced by running `plot_high_level_figures.py` with the saved run directory.

Table 4: Wind-speed RMSE by observed regime and changes after member removal with Inner-Val weight recalibration. Wind RMSE and its changes are in mph. Aggregate MSE uses all six standardized outputs. Positive changes indicate greater error relative to the frozen ensemble.

Regime	N	Transformer	iTransformer	TimesNet	Ensemble
Low-wind	227	0.3309	0.3866	0.4343	0.3372
Moderate-change	83	0.6010	0.7438	0.5848	0.5526
High-ramp	132	1.2235	1.2537	1.2972	1.2076

			Change in wind RMSE (mph)		
Removed	MSE	ΔMSE	Low-wind	Moderate-change	High-ramp
None (full ensemble)	0.203396	0.000000	0	0	0
Transformer	0.214920	+0.011524	+0.05602	+0.05149	+0.02635
iTransformer	0.203996	+0.000600	+0.00010	-0.00774	+0.00856
TimesNet	0.204531	+0.001135	-0.00103	+0.02370	-0.01086

Appendix A. Controlled supplementary experiments

The supplementary comparisons change one aspect of the iTransformer input while preserving the six forecast outputs. They use the same hourly record, three chronological data boundaries, one-hour target horizon, and checkpoint selection rule as the main comparison. Aggregate errors therefore refer to a common set of standardized targets within each experiment. The purpose of these runs is to inspect the direction and size of the sensitivity under a defined architecture, rather than to retune the ensemble against the test block.

Appendix A.1. Feature ablation

The full-input arm includes the nine candidate covariates and wind speed. The screened arm retains the five leading MI covariates plus wind speed. In both arms, the output channels are the same six variables, so the inclusion of extra inputs in the full-input arm does not increase the number of targets in its loss. Both use 24 hours of history and the same 1,036 fitting timestamps. Standardization uses the fitting block, and the best checkpoint is selected by Outer-Val MSE. These controls are particularly important in a multivariate forecaster: including the additional covariates in the loss as extra outputs would change the task, making an aggregate MSE comparison uninformative.

Table A.5: iTransformer feature ablation with matched fitting and target timestamps. Wind MSE is in mph^2; aggregate errors use standardized channels.

Inputs	Count	Outer-Val MSE	Test MSE	Test MAE	Mean R^2	Wind MSE
All features	10	0.201924	0.243347	0.280899	0.694338	0.674579
MI Top-5 + wind	6	0.195749	0.240319	0.277415	0.699330	0.650078

Table A.5 and Fig. A.5 show that screening reduces standardized test MSE by 1.24% and wind-speed MSE by 3.63%. Wind-speed R^2 increases from 0.610220 to 0.624377. The screened arm also has lower validation MSE. The checkpoints occurred at epochs 11 and 7 for the full and screened arms, respectively. With fewer tokens, iTransformer can allocate its representation to the covariates most strongly associated with wind speed on the fitting block. Removing redundant inputs may also reduce the number of relationships that must be estimated from the available windows. The observed gains are consistent with those mechanisms, although the comparison combines a change in input dimensionality with a change in the identity of the inputs.

The contrast between the two arms jointly varies input dimensionality and covariate composition. Either change may contribute to the observed difference, and the present design cannot identify their separate effects. The experiment evaluates a single subset and training seed, so the result pertains to this implementation of iTransformer and this chronological split. An excluded variable could provide useful information under different meteorological conditions.

A matched-dimensionality control would retain wind speed and select five additional covariates using absolute Pearson correlation estimated on the fitting block. A second control would evaluate several prespecified random five-covariate subsets, with wind speed retained in each. Both comparisons would hold the forecast targets, model architecture, checkpoint

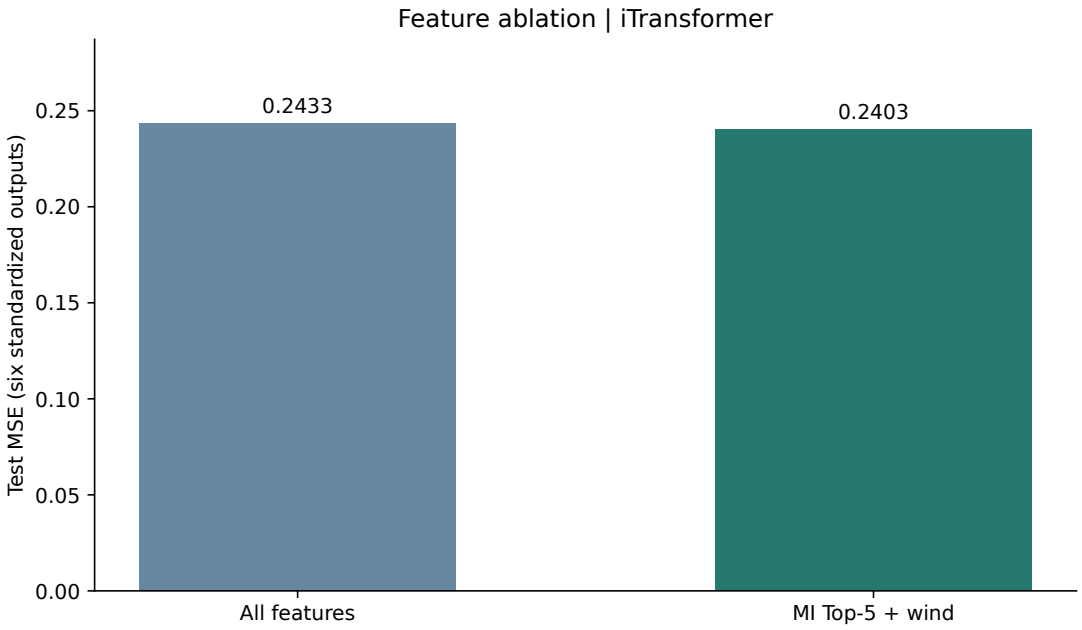


Figure A.5: Standardized test MSE for iTransformer with all candidate inputs or the MI-screened inputs. Both arms predict the same six output channels.

criterion, and evaluation timestamps fixed. Repeating the comparison across training seeds and chronological splits would help determine whether the MI ranking contributes predictive value beyond that associated with a smaller input set.

Appendix A.2. Sensitivity to historical window length

Four iTransformer arms use 12, 24, 48, and 96 hours of context. Each uses the screened input set and trains on the same 964 target timestamps. The earliest eligible fitting target is at index 96, which gives even the longest-history arm a full causal context. With different fitting timestamps, a long-history arm would also be deprived of more early training examples. Restricting all arms to the common target support removes that source of variation. Outer-Val and test target timestamps are likewise matched. The 24-hour sensitivity arm uses fewer fitting examples than the 24-hour feature ablation and main experiment, which explains why its loss need not equal either of their losses.

Table A.6: History-length sensitivity with identical fitting target support. Boldface identifies the lowest Outer-Val MSE, which determines the selected length.

Hours	Outer-Val MSE	Test MSE	Test MAE	Mean R^2	Wind MSE (mph^2)
12	0.230234	0.291384	0.281010	0.628838	0.684056
24	**0.195590**	0.244623	0.272974	0.693470	0.656530
48	0.209198	0.241446	0.287952	0.693329	0.657773
96	0.201672	0.239448	0.293980	0.695422	0.674883

The 24-hour arm attains the lowest Outer-Val MSE (Table A.6 and Fig. A.6). It also has the lowest test aggregate MAE and wind-speed MSE among these candidates. The 96-hour

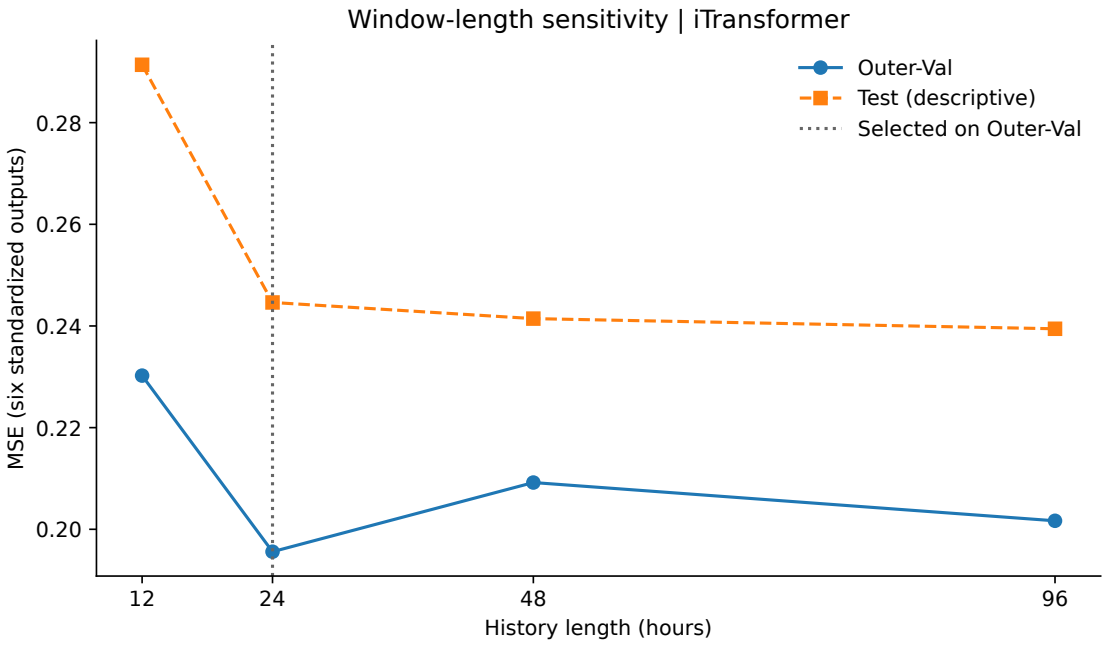


Figure A.6: Validation and test aggregate MSE as a function of iTransformer history length. Outer-Val MSE favors 24 hours, while aggregate test MSE favors 96 hours on this split.

arm achieves the lowest test aggregate MSE. Consequently, the window experiment supports 24 hours under the validation criterion and for some test metrics, while showing that the rank depends on the loss and time block. The test comparison is reported as a diagnostic; it does not change the history length of the evaluated ensemble.

A 12-hour context can miss the previous occurrence of daily variability. A 24-hour context can retain that information with shorter input representations and less dependence on several days of past conditions. Longer contexts can help when low-frequency patterns persist, and can also bring observations from an earlier weather regime into the input. The difference between 24 and 96 hours is therefore a trade-off between temporal context and the relevance of older measurements. The sensitivity experiment tests the forecast consequences of this trade-off on the available data. Establishing the mechanism would require multiple seasons and explicit controls for periodicity and regime changes. The current test also changes the number of input coordinates and associated embedding parameters, which should be considered when comparing compute or capacity across history lengths.

Appendix B. Supplementary stability diagnostics

Appendix B.1. Calibration weights across retraining runs

Table B.7 reports the active weights in percent. PatchTST, LSTM, and DLinear are effectively zero in all three runs. Autoformer enters with a share below half a percent in the latter two runs. Reporting near-zero entries at a fixed precision avoids interpreting numerical residuals on the simplex boundary as substantive model participation.

The mass on Transformer varies by less than four percentage points, whereas the distribution between iTransformer and TimesNet changes more. Weight sensitivity and loss

Table B.7: Frozen Inner-Val ensemble weights across seeds. Values are percentages rounded independently.

Seed	Transformer	iTransformer	TimesNet	Autoformer
2021	51.83	24.83	23.34	0.00
42	55.78	16.74	27.37	0.11
2023	53.07	26.75	19.75	0.43

sensitivity need not track one another because the objective is a quadratic form in the residual matrix. A small eigenvalue in a feasible direction allows the weights to move along that direction with little change in calibration loss. Directly comparing weight stability therefore requires the aligned residual covariance or the forecasts themselves. The current evidence is strongest at the level of the three main contributors and of aggregate score dispersion.

Appendix B.2. Paired temporal diagnostics

Table B.8: Four contiguous test periods. Wind MAPE (%) is rounded directly from the saved full-precision metrics. Times are those recorded in the dataset, in 2024.

Period	Start	End	N	SLSQP	Transformer	Equal
1	Jan 13 15:00	Jan 18 05:00	111	20.57	21.31	22.80
2	Jan 18 06:00	Jan 22 20:00	111	22.34	22.63	24.12
3	Jan 22 21:00	Jan 27 10:00	110	22.67	22.47	26.53
4	Jan 27 11:00	Feb 1 00:00	110	20.79	20.71	21.42

	Observed wind (mph)		Wind MSE (mph^2)		SLSQP
Period	Mean	RMS hourly change	SLSQP	Transformer	Lower AE (%)
1	2.54	0.87	0.487	0.456	58.6
2	2.51	0.88	0.570	0.592	53.2
3	2.25	0.80	0.467	0.492	49.1
4	2.75	1.02	0.682	0.745	43.6

The RMS change uses consecutive observations within each block. The final column reports the fraction of timestamps at which the SLSQP absolute error is strictly smaller than Transformer's. For positive observations, the sign of the absolute percentage error difference at a given timestamp equals the sign of the absolute error difference. The corresponding period averages, however, depend on the size of each error difference and the observed speed used as its denominator. This explains how the method that is better at fewer origins can still produce lower squared loss across a block.

Appendix B.3. A squared-error identity for complementary forecasts

For a single target and any feasible convex weights, the squared ensemble residual admits the identity

$$(\hat{y}_{\text{ens}} - y)^2 = \sum_m w_m(\hat{y}_m - y)^2 - \sum_m w_m(\hat{y}_m - \hat{y}_{\text{ens}})^2. \tag{B.1}$$

To see this, expand each squared residual around $\hat{y}_{\text{ens}}$. The cross term vanishes because $\sum_m w_m(\hat{y}_m - \hat{y}_{\text{ens}}) = 0$. Averaging over time preserves the equality. The second term measures forecast disagreement and quantifies the reduction from the weighted mean of constituent squared errors. The weighted mean of the constituent losses can itself exceed that of the best member, so disagreement alone is insufficient to beat that member.

For the wind channel, the mean disagreement term is 0.065, 0.043, 0.046, and 0.091 mph^2 across the four blocks. In Period 4, adding 0.091 to the ensemble MSE of 0.682 gives a weighted mean constituent MSE of approximately 0.773. That value is above Transformer's 0.745, yet the disagreement term brings the combination below Transformer. In Period 1, the disagreement term is smaller than the gap between the weighted mean constituent MSE and the Transformer MSE; consequently Transformer has the lower wind MSE for that block. For the three materially weighted models, test wind residual correlations range from 0.76 to 0.86. The models share substantial error structure, leaving a limited but measurable opportunity for cancellation. This identity concerns squared loss; absolute percentage loss can rank the same predictions differently.

Appendix C. Reproduction and audit details

Appendix C.1. Forecast-origin information and artifact lineage

For each target index, the origin is one hour earlier. Model inputs contain the preceding recorded observations, scaled using the fitting-block mean and standard deviation. Calendar features use known timestamps. The model cannot consume the target's realized wind speed or realized meteorological covariates. For a window that crosses a split boundary, the preceding observations are legitimate inputs because they occur before the forecast origin. The identity of the target block is determined by the label index and is independent of the location of the first observation in the window.

A reproducible run records the range of fitting rows, the input feature order, the per-channel scaler, the forecast horizon, and the timestamps used for checkpoint selection and calibration. An aligned prediction array has one first-axis entry per target timestamp, one second-axis entry per horizon step, and one last-axis entry per forecast channel. The saved run uses one horizon step. Before comparing or combining model outputs, both the shape and label array must match across models. Shape equality alone cannot detect a one-hour shift or a different channel order. Such shifts are particularly consequential for percentage errors around steep changes in wind speed.

The point-forecast artifacts include the base checkpoints, Inner-Val predictions and labels, frozen weights, and test predictions and labels. They allow recovery of both the shared coefficient vector and the output-scale conventions. The supplemental artifacts record the metric values for the feature and window arms, per-seed frozen weights, and the period boundaries. When recomputing an error from a saved prediction array, the mean and scale should be taken from the same run. Mixing scaler parameters across splits can distort physical-unit errors even when the standardized arrays have the same shape.

Appendix C.2. Preprocessing assumptions and test access

The analysis begins with the supplied cleaned hourly record. The forecasting pipeline evaluated here performs no missing-value imputation or interpolation. In particular, it does

not use observations from later timestamps to fill earlier values. The upstream missing-value treatment could not be verified from the supplied archive. If the upstream process used two-sided interpolation or another operation requiring later observations, the resulting values might include information that would have been unavailable at the forecast origin. The chronological safeguards described here apply to the computations performed after receipt of the cleaned record. An operational reproduction should retain raw observation times and quality flags, document how missingness is handled, and reproduce a causal alternative where required. The archive used here does not establish station metadata, measurement height, or full upstream cleaning provenance. Those details are necessary to extend the interpretation to hub-height wind or to compare physical-unit errors across locations.

The existing test timestamps also overlap earlier exploratory analyses. The revised chronology prevents direct fitting of the reported scaler, ensemble weights, and checkpoint choice on the designated test labels. Evidence from the revised run remains conditional on the project history. A prospective evaluation on a new chronological block would isolate the final design from any indirect influence of the earlier exploratory results. The held-out metrics and supplementary comparisons should consequently be interpreted as a retrospective evaluation of the documented protocol.

Appendix C.3. Replication inventory

Table C.9: Artifacts needed to reproduce the numerical comparisons. Paths are relative to the project root.

Item	Project artifact
Chronology, scaler, and model setup	`nested_60_20_20_run/protocol.json` and per-model `training.json`
Calibrated combination	`nested_60_20_20_run/frozen_weights.json` and per-model Inner-Val prediction arrays
Main aggregate and wind scores	`nested_60_20_20_run/metrics.json` and per-model test prediction arrays
Adaptive uncertainty analysis	`plot_high_level_figures.py` and the principal run directory
Feature and window comparisons	`additional_experiments_run/ablation/metrics.csv` and `additional_experiments_run/sensitivity/metrics.csv`
Training repeatability	`additional_experiments_run/seeds/seed_metrics.csv` and the three seed run directories
Time-block comparison	`additional_experiments_run/periods/period_metrics.csv`

For the reported adaptive bounds, resampling starts in the residual calibration block and uses 2,000 replicates of 12-hour blocks. The resampled quantile means are applied on the wind-speed scale after multiplying by the local volatility. The quantile offsets remain fixed during the test, but the local scale is recomputed from the prior observed window at each origin. Because of truncation at zero, the final width can be smaller than the untruncated interval at very low predicted wind speeds. A comparison with an alternative interval method

should use the same origin times, valid targets, normalization range, and nominal coverage before attributing a difference to calibration.

Appendix D. Additional data and uncertainty diagnostics

Appendix D.1. Wind-speed distributions across chronological blocks

Table D.10 describes the recorded wind speed before standardization. The fitting block has the highest mean and largest standard deviation. The mean decreases from 2.89 mph in fitting to 2.51 mph in test, while the test standard deviation is 1.32 mph. The largest RMS hourly change among the four blocks occurs in Outer-Val even though the fitting block spans the broadest observed range. Thus, spread across wind speeds and typical successive changes capture different aspects of the forecasting problem. A period may have a broad range because daily peaks are high, while a different period can have larger hour-to-hour changes around a smaller range of speeds.

Table D.10: Descriptive wind-speed statistics on the recorded physical scale. The variance uses the full block size as its denominator, and hourly changes are computed within each block. All statistics other than the observation count are in mph.

Block	Observations	Mean	SD	Minimum	Maximum	RMS change
Fitting	1,060	2.892	1.487	1.0	9.3	0.845
Inner-Val	264	2.613	1.279	1.0	7.0	0.899
Outer-Val	442	2.800	1.407	1.0	8.2	0.925
Test	442	2.514	1.316	1.0	8.4	0.895

The shifts are relevant to both point and interval calibration. For point forecasts, squared error in standardized units uses a fixed scale from fitting, so a change in test dispersion does not silently rescale the reported MSE. The denominator of R^2, in contrast, uses the dispersion of the evaluation observations. Consequently, scores from different test records may diverge even at equal physical-unit MSE. For interval forecasting, similar RMS hourly changes in Inner-Val and test make volatility scaling a plausible starting point. Comparable aggregate volatility does not imply identical conditional residual tails. Large ramps and near-calm conditions can occur at different frequencies, and a trailing RMS statistic cannot distinguish all physically distinct regimes.

Appendix D.2. Skill relative to hourly persistence

Hourly persistence predicts $\hat{y}_t = y_{t-1}$, where y_{t-1} is the last observation available at the forecast origin. It uses no fitted coefficients and is evaluated at all 442 test targets, including the first target, for which the prior observed speed is at the last Outer-Val timestamp. Its test wind MSE, MAE, and MAPE are 0.799312 mph^2, 0.606109 mph, and 24.17%, respectively. The principal SLSQP ensemble yields 0.551304 mph^2, 0.521900 mph, and 21.59% under the same wind-only comparison. Its MSE is approximately 31.0% lower than persistence. Because this reference predicts wind alone, it has no directly comparable six-output MSE.

This comparison shows that the ensemble improves on carrying the previous wind observation forward under the documented horizon. Persistence can still be competitive over

stable episodes, especially in low-variability periods. Its error is exactly the next hourly wind increment, so its performance reflects how frequently and sharply the observed process changes. A stronger operational benchmark suite would include seasonal persistence, a short autoregression trained on the fitting block, and an NWP forecast from the same issuance time. Those comparisons would help determine whether the added computation of a heterogeneous ensemble is warranted for a particular scheduling application.

Appendix D.3. Interval coverage and miss direction by period

Using the original 12-hour bootstrap configuration, we recomputed pointwise coverage and width on the same four periods as the point comparison. Table D.11 reports lower and upper misses separately. Pooled across periods, 13 observations fall below the lower bound and six exceed the upper bound. The counts indicate the direction of the realized misses under the fitted offsets. With 19 total misses on a serially dependent sequence, the difference does not establish that a general recalibration of the lower tail is warranted. A wind-farm operator may assign different costs to each direction, which would call for a pre-specified tail or decision loss.

Table D.11: Nominal 95% adaptive interval diagnostics by test period. The bounds are those of the principal 12-hour bootstrap configuration.

Period	Targets	Below lower	Above upper	PICP (%)	MPIW (mph)
1	111	5	1	94.59	3.602
2	111	2	2	96.40	3.671
3	110	3	1	96.36	3.574
4	110	3	2	95.45	3.925
All	442	13	6	95.70	3.693

Period 4 has the largest average interval width and the largest RMS hourly wind change, while Period 3 has the smallest mean width and smallest hourly change. These two blocks illustrate that the trailing scale responds to differences in observed variability, although the width statistic also depends on clipping and the distribution of volatility within a block. Period 1 has the lowest coverage and the most lower-bound misses despite having a mean width larger than that of Period 3. The comparison illustrates why a width diagnostic is most informative when paired with the realized coverage and miss direction. Assessing clustered failures would additionally require measuring the persistence of misses and the variability of the underlying forecasts during the affected hours.

Appendix D.4. Exploratory sensitivity to bootstrap block length

The 12-hour trailing volatility window is a feature of the scale model. The bootstrap block length, by contrast, is the number of consecutive standardized calibration residuals sampled together. The main experiment uses 12 hours for both. To separate the two roles, we held the local scale fixed and recomputed the bootstrap offsets at block lengths of 6, 12, and 24 residuals. Each configuration uses 2,000 replicates and seed 42. We keep the point forecasts and 264 Inner-Val residuals fixed. Only the resampling of residuals changes, and the bounds are assessed at the same 442 timestamps.

Table D.12: Exploratory interval sensitivity to bootstrap block length. The 12-hour volatility window and all point forecasts are held fixed.

Block length	q_L	q_U	PICP (%)	MPIW (mph)	PINAW
6	−1.9229	2.6619	95.70	3.6410	0.4920
12	−1.9255	2.7229	95.70	3.6926	0.4990
24	−1.9397	2.8661	95.70	3.8186	0.5160

The alternative lengths produce the same observed coverage count and different mean widths. Increasing the block length from 6 to 24 increases MPIW by approximately 0.18 mph, driven chiefly by an increase in the upper residual offset. Equal coverage counts can occur if the interval boundaries move without crossing an additional observation, so this result should not be interpreted as identical calibration at every forecast origin. It also does not imply that the shortest block is optimal. Short blocks preserve less residual history, while longer blocks reduce the number of independently drawn segments within each replicate and can give individual calibration episodes greater influence. We retain the 12-hour result as the principal configuration and use this comparison as a descriptive sensitivity check.

The sensitivity analysis varies the bootstrap block length while retaining adaptive scaling in every configuration. The study has yet to evaluate a constant-width residual interval. Consequently, the reported coverage and width do not establish an advantage attributable to the adaptive scale. In future work, a constant-width baseline will add fixed lower and upper residual quantiles to the same point forecasts, with the quantiles estimated from an independent calibration block or chronologically out-of-sample residuals. For a strictly constant width on the reported scale, any adjustment at zero would shift both bounds together. A paired evaluation at the same test origins and nominal coverage would compare empirical coverage, mean width, and interval score, both overall and within prespecified volatility strata.

A longer independent record would also support a comparison of block lengths under the same calibration and evaluation protocol. A proper interval score can complement coverage and width by explicitly penalizing the magnitude of misses. For a central $(1 - \alpha)$ interval $[L_t, U_t]$, the interval score is

$$\mathrm{IS}_\alpha(L_t, U_t; y_t) = (U_t - L_t) + \frac{2}{\alpha}(L_t - y_t)\mathbf{1}\{y_t < L_t\} + \frac{2}{\alpha}(y_t - U_t)\mathbf{1}\{y_t > U_t\}. \tag{D.1}$$

The score distinguishes two intervals with the same width and coverage count when one misses observations by larger margins. Such a comparison is particularly relevant for dispatch, where the severity of a tail miss can matter more than its occurrence alone. A downstream evaluation could replace the symmetric penalty with an operational cost that reflects shortfall and surplus separately.

Appendix D.5. End-to-End Inference Cost

We benchmarked the saved checkpoints on an NVIDIA GeForce RTX 3060 Laptop GPU with PyTorch 2.7.1 and CUDA 11.8. Each configuration used float32 and batch size one. We warmed up with 20 calls and timed one forecast at each of 100 origins spaced across

the test block. CUDA synchronization brackets the timed forward passes. The forward measurement begins with prepared input tensors on the GPU and includes the weighted sum for the ensemble. The local pipeline starts from buffered observations and timestamps, and includes window and calendar-feature preparation, scaling, device transfer, forward inference, inverse scaling, and the adaptive interval for the ensemble. Model loading, training, weight fitting, and bootstrap quantile calibration occur outside the timed path. Peak memory is the maximum CUDA tensor memory allocated by PyTorch, including resident model parameters; it excludes allocator-reserved space and memory used by other processes.

Table D.13: Batch-one forecast latency and peak GPU tensor allocation for the frozen checkpoints. The local ensemble path also produces the wind-speed interval.

	Forward (ms)		Local pipeline (ms)		Peak memory
Model	P50	P99	P50	P99	(MiB)
Transformer	2.62	4.06	3.42	5.23	69.18
Ensemble	16.39	23.58	17.54	25.74	133.93

The local ensemble pipeline had a P99 latency of 25.74 ms on this device, several orders of magnitude below the one-hour forecast horizon. The result establishes the computational headroom of the measured single-stream pipeline under the benchmark conditions. Operational real-time performance depends on the actual issuance deadline, telemetry arrival and validation, service scheduling, concurrent sites, and delivery to the dispatch system. The benchmark does not include these factors or validate a dispatch policy. The 100 timed origins also provide limited resolution for estimating tail latency.

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